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Aircraft MRO 2.0: The digital revolution

A new survey suggests AI-powered advances could reshape aircraft maintenance, repair, and overhaul. But companies will need to embrace digital disruption.

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For airlines around the world, fleet maintenance is increasingly challenging. In the United States, for instance, airlines have endured a 15 percent increase in maintenance costs over the past five years. Meanwhile, there's been a 14 percent increase in the share of flights delayed. This has raised the pressure on already thin margins in a competitive industry.

Although maintenance, repair, and overhaul (MRO) organizations endeavor to provide efficient aircraft maintenance while controlling costs, traditional levers such as lean management and enhanced scheduling are becoming insufficient amid a shifting industry landscape. New approaches are needed.

MROs may accelerate improvements through AI-powered solutions that have the potential to enable better performance—helping, for example, to predict maintenance needs so proper labor-and-materials resources will be on hand and then sequencing tasks for maximum efficiency. To seize this digital advantage, MROs should become comfortable with adopting technology that could disrupt the status quo. Some might be weary of hype about digital capabilities, feeling that associated results did not materialize in the past, but more than ever it is now possible to capitalize on modern advancements in technology and analytics.

To understand the shifting digital environment within aerospace MROs, we conducted a global survey of the industry in 2023, asking 45 executives and managers at MRO organizations around the world about the current state of and future aspirations for their companies' digital efforts. These findings show that while MROs are bullish on the impact that digital might have on their operations, they need to undertake fundamental digital transformations in order to fully realize those gains. They might not know how to start, feel they lack the in-house technology skills to execute on digital priorities, fear that their largely paper-based industry does not generate enough digital data to make a transformation worthwhile, or all of the above. But the survey results also suggest that making digital development a priority—while injecting fresh talent who can help execute on digital's promise—could offer significant benefits and aid MRO organizations in their efforts to push the industry forward.

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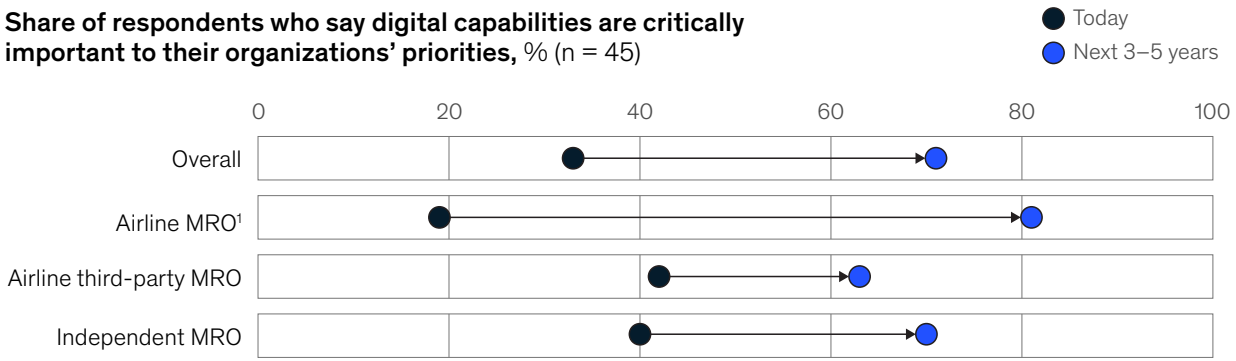
The next three to five years will be pivotal for digital adoption

While only 33 percent of respondents say digital is critically important to achieving their organizations' priorities and objectives today, more than 70 percent say that it will be critically important within three to five years. This pattern suggests that many organizations have not yet begun their digital journeys.

Attitudes about digital adoption also varied among the MRO segments represented in our survey: airline-affiliated MROs that provide services to third parties, airlines that maintain their own fleets, and independent MROs that provide services to OEMs and other third parties. Notably, respondents from airline MROs are 25 percent more likely than their industry peers to believe that digital will be critical in the implementation of future strategy. Because their airline parent companies are consumer-facing and tend to be comparatively tech-forward, airline MROs might be more likely than their peers to have witnessed the benefits of digital transformation firsthand.

Most maintenance, repair, and overhaul organizations say digital capabilities will be critical in coming years.

Share of respondents who say digital capabilities are critically important to their organizations' priorities, % (n = 45)



[1]Manufacturing, repair, and overhaul.
Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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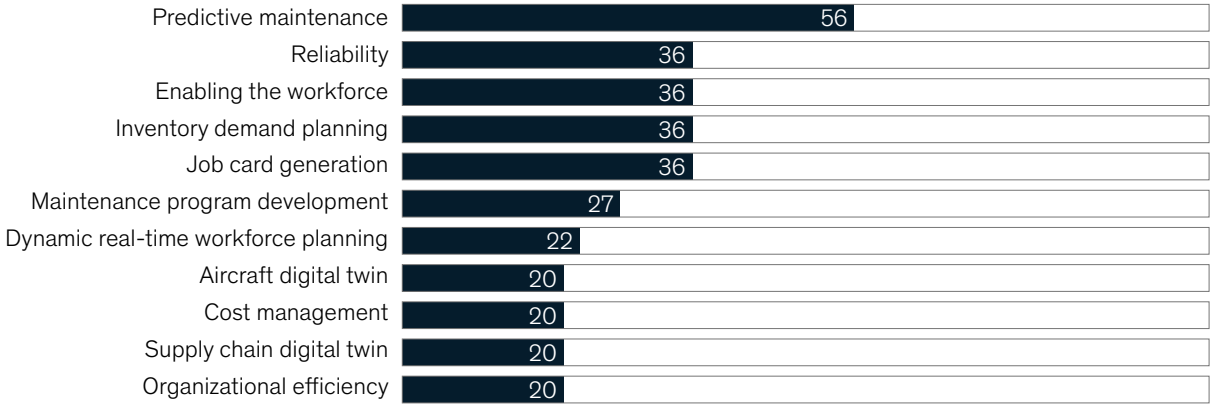
Predictive maintenance is a clear digital priority for MROs

When we asked respondents to identify what their organizations' top digital priorities will be over the next three to five years, 56 percent of them named predictive maintenance—identifying and planning for nonroutine maintenance issues in advance—as one of those priorities. Given that up to 60 percent of technicians' time can be spent on unplanned work (often on problems that are only discovered in the course of remedying other problems), it makes sense that MRO respondents would be eager to take advantage of digital's ability to flag potential maintenance issues before they arise.

Predictive maintenance is consistently cited as a digital priority by respondents from airline MROs, airline-affiliated MROs, and independent MROs. Other priorities that appeared in the top three results for all MRO categories include reliability, which is the likelihood that components will perform as intended, and AI job card generation, which involves the automated creation of detailed, optimally sequenced maintenance instructions. When we asked executives where advanced digital solutions are being deployed today, predictive maintenance, reliability, and job card generation also appeared in the top five results.

Predictive maintenance was the top digital priority for maintenance, repair, and overhaul survey respondents.

Organizations' top digital priorities for maintenance, repair, and overhaul in the next 3–5 years,
% of respondents (n = 45)



Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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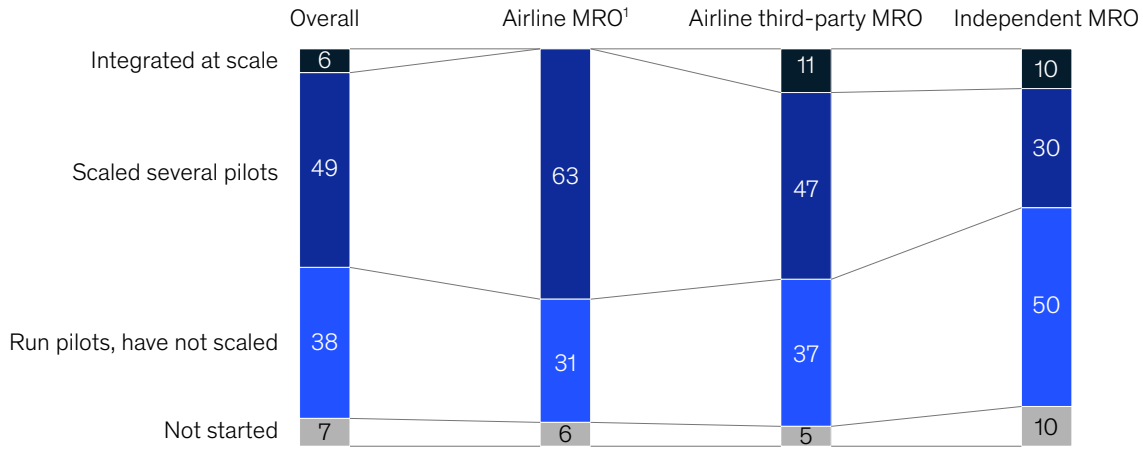
Only 6 percent of MROs have integrated digital at scale

Roughly half of respondents say their MRO organizations have scaled multiple digital pilots—for example, experimenting with the use of tablets in frontline maintenance. But only 6 percent believe their organizations have gone further by integrating digital and analytics at scale throughout the organization to enhance engineering, the supply chain, quality assurance, and other areas.

This trend is not unique to MROs—digital transformations are challenging in any industry. McKinsey research shows, in fact, that only 16 percent of attempted digital transformations successfully enabled organizations to sustain change over the long term. When the focus is on more traditional industries that share traits with aviation MROs, such as automotive and infrastructure, success rates are even worse at between 4 and 11 percent.

Very few maintenance, repair, and overhaul organizations have implemented digital solutions at scale.

Organizations' implementation of digital and analytics solutions, % of respondents (n = 45)



[1]Manufacturing, repair, and overhaul.
Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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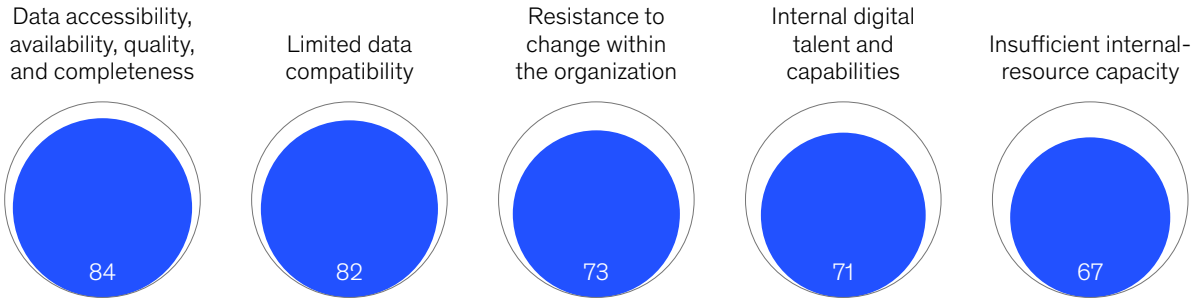
Data limitations and resistance to change are among the major barriers to overcome

More than 80 percent of respondents consider data limitations to be the most significant barrier to digital adoption at their organizations. This sentiment could result from the MRO industry's reliance on physical recordkeeping (in large part due to regulatory mandates requiring paper records), which can create fragmented data collections that are difficult to transfer into digital databases. Machine learning solutions—which can generate insights from unstructured, partial source data—could eliminate many of these quality issues.

Organizational resistance to change and the lack of internal digital talent are both cited by more than 70 percent of respondents as additional barriers to successful deployment and scaling of digital capabilities.

Maintenance, repair, and overhaul organizations cited data issues and resistance to change as the greatest barriers to scaling digital solutions.

Biggest barriers to successfully deploying advanced digital solutions, % of respondents (n = 45)



Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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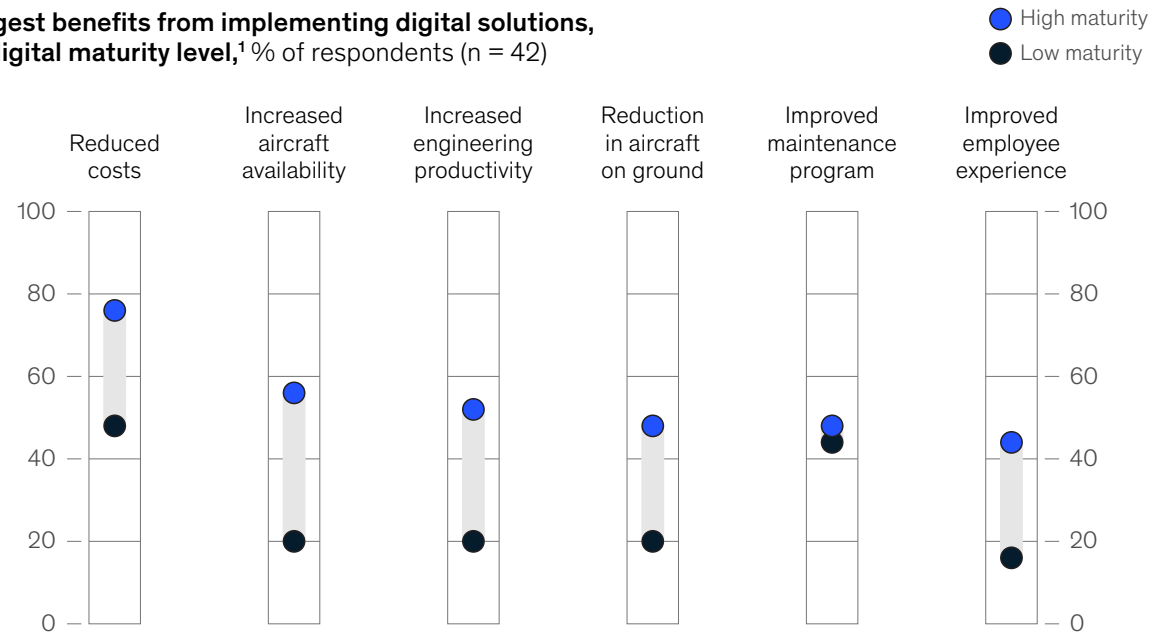
Front-runners are seeing the benefits in revenue and productivity

Respondents whose organizations have embraced digital say that the advantages are clear across the value chain. Fifty percent of these respondents say they've experienced revenue uplifts of more than 5 percent and a more than 10 percent improvement in engineering productivity. More than 25 percent of respondents indicate that they have realized a 10 to 20 percent reduction (or more) in maintenance spending as a result of digital efforts.

Maintenance, repair, and overhaul organizations with high digital maturity have achieved improvements in multiple areas.

Biggest benefits from implementing digital solutions, by digital maturity level,[1] % of respondents (n = 42)



[1]Low maturity = respondents who indicated their organizations have not started implementing digital or scaling pilots (n = 17); high maturity = respondents who indicated their organizations have fully implemented digital or scaled multiple pilots (n = 25).
Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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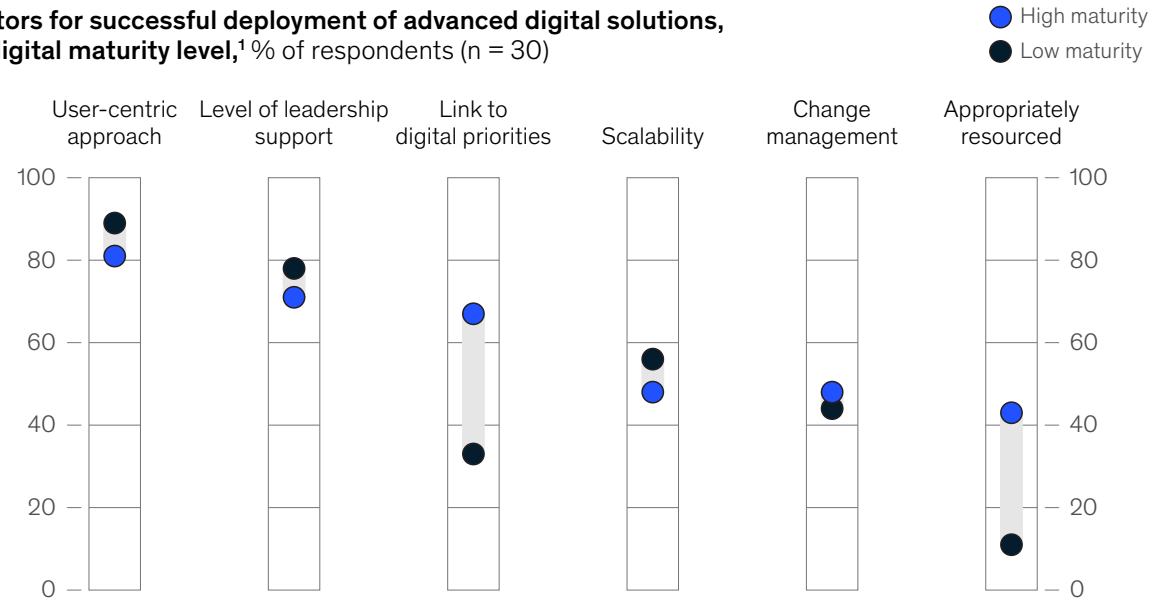
Success flows from a user-centric approach

Respondents whose organizations have successfully integrated digital and analytics at scale primarily credit their success to three factors:

- adopting a user-centric approach (in some cases by cocreating digital solutions with end users such as technicians and shop floor managers)
- securing strong leadership support and endorsement of the initiative
- closely aligning solutions with the organization's broader digital priorities

Survey respondents noted that a user-centric approach was critical to deploying advanced digital solutions.

Factors for successful deployment of advanced digital solutions, by digital maturity level,[1] % of respondents (n = 30)



[1]Low maturity = respondents who indicated their organizations have not started implementing digital or scaling pilots (n = 9); high maturity = respondents who indicated their organizations have fully implemented digital or scaled multiple pilots (n = 21).
Source: McKinsey MRO Digital Survey, Sept 11–21, 2023

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Only a limited number of MRO organizations have thus far integrated digital solutions at scale, but the benefits for those that have are evident in these survey results. The survey also indicates a clear consensus that digital advances will be crucial in achieving new levels of business and operational performance over the next five years. The following steps can help MRO organizations start down the path to digital integration:

— aligning senior leadership on a focused digital aspiration—with an accompanying road map that is closely tied to achieving value

— identifying a technology leader to attract talent who can unite tech, business, and operations in a reimagined, integrated enterprise

— institutionalizing governance of the transformation, putting a team in charge that can reinforce priorities, establishing a rigorous cadence, and holding the organization accountable for achieving planned outcomes

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Realizing the full value of modern digital solutions could require MROs to inject new talent into the organization, invest in new capabilities, and embrace a new way of working. As the industry faces evolving challenges such as supply chain complexity, talent shortages, and rising competitive pressures, MRO organizations that can add advanced digital solutions to their tool kits could be well positioned to succeed.

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